



**Linear Programming (LP): A mathematical method for finding the best possible outcome (maximum profit or minimum cost) when there are limitations.**

### **TERMS ASSOCIATED WITH LINEAR PROGRAMMING**

**Decision Variables:** The things we can control or choose (e.g., number of products to make).

**Objective Function:** The formula we want to maximize (profit) or minimize (cost).

**Constraints:** The limitations or restrictions on resources (time, money, materials).

**Feasible Region:** The set of all possible solutions that satisfy all constraints.

**Optimal Solution:** The best solution (highest profit or lowest cost) within the feasible region.

**Non-negativity:** The rule that you cannot produce negative quantities ( $x \geq 0$ ,  $y \geq 0$ ).



| Term                           | Meaning                                                                       |
|--------------------------------|-------------------------------------------------------------------------------|
| <b>Linear Programming (LP)</b> | Mathematical method for best outcome (max profit/min cost) under constraints. |
| <b>Decision Variables</b>      | Quantities we control: $x$ = number of chairs, $y$ = number of tables.        |
| <b>Objective Function</b>      | Formula to maximize (profit) or minimize (cost).                              |
| <b>Constraints</b>             | Limitations: time, materials, budget (inequalities).                          |
| <b>Feasible Region</b>         | Set of all solutions satisfying constraints.                                  |
| <b>Optimal Solution</b>        | Best feasible solution (maximum or minimum).                                  |
| <b>Non-negativity</b>          | $x \geq 0, y \geq 0$ (no negative production).                                |

## A. WHAT IS LINEAR PROGRAMMING? (Definition)

**Linear Programming (LP) is a mathematical method used to find the best possible outcome (maximum profit, minimum cost, etc.) in a given situation with linear relationships.**

**Simple meaning: How to get the best result with limited resources.**

### Why "Linear"?

**All relationships are straight-line equations or inequalities. Variables appear only to the first power (no  $x^2$ , no  $xy$ , no  $\sqrt{x}$ ).**

### Examples:

✓ **Linear:  $2x + 3y \leq 100$**

✓ **Linear:  $x + y = 50$**



✗ **Non-linear:**  $x^2 + y \leq 100$  (has  $x^2$ )

✗ **Non-linear:**  $xy = 50$  (has  $xy$ )

✗ **Non-linear:**  $\sqrt{x} + y \geq 20$  (has  $\sqrt{x}$ )

## B. KEY COMPONENTS OF LINEAR PROGRAMMING

| Component             | Meaning                          | Example                                |
|-----------------------|----------------------------------|----------------------------------------|
| 1. Decision Variables | What we choose/control           | $x = \text{chairs}, y = \text{tables}$ |
| 2. Objective Function | Maximize profit or minimize cost | $\text{Max } P = 50x + 40y$            |
| 3. Constraints        | Limitations (inequalities)       | $2x + y \leq 100$ (time limit)         |
| 4. Non-negativity     | Cannot produce negative amounts  | $x \geq 0, y \geq 0$                   |
| 5. Feasible Region    | All possible solutions           | Shaded area on graph                   |
| 6. Optimal Solution   | Best solution (max/min)          | Corner point of feasible region        |

**Memory aid:** "Decide Objective, Check Non-negativity, Find Optimal" (DO CN FO)

**Memory Aid: "Decide Objective, Check Non-negativity, Find Optimal" (DO CN FO)**

## C. STEPS TO FORMULATE AN LP PROBLEM (From Word Problem)

### Step 1: Define Decision Variables

→ **What quantities can we choose or control?**

→ **Let  $x = \dots$ ,  $y = \dots$**

### Step 2: Write the Objective Function



- **What are we trying to maximize or minimize?**
- **Maximize (or Minimize)  $Z = \dots$**

### **Step 3: Identify Constraints**

- **What are the limitations or restrictions?**
- **List all inequalities**

### **Step 4: Add Non-negativity Constraints**

- **Usually:  $x \geq 0, y \geq 0$  (cannot produce negative quantities)**

### **Step 5: Summarize the Complete Formulation**

- **Present all parts together**

### **EXAMPLE OF FORMULATION STEPS**

**Problem: A factory produces two types of chairs: dining chairs and office chairs.**

**Given:**

**Dining chair: 2 hours labor, 3 kg wood, profit ₦5,000**

**Office chair: 3 hours labor, 2 kg wood, profit ₦6,000**

**Available: 120 hours labor, 150 kg wood**



**Step 1: Define Decision Variables**

**Let  $x$  = number of dining chairs produced per week**

**Let  $y$  = number of office chairs produced per week**

**Step 2: Write Objective Function**

**Maximize Profit:  $P = 5000x + 6000y$**

**Step 3: Identify Constraints**

**Labor constraint:  $2x + 3y \leq 120$**

**Wood constraint:  $3x + 2y \leq 150$**

**Step 4: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 5: Complete Formulation**

**Maximize  $P = 5000x + 6000y$**

**Subject to:**

**$2x + 3y \leq 120$**

**$3x + 2y \leq 150$**

**$x \geq 0, y \geq 0$**



**D. REAL-LIFE APPLICATIONS OF LINEAR PROGRAMMING**



| Area             | Example                                                                            |
|------------------|------------------------------------------------------------------------------------|
| Manufacturing    | Product mix to maximize profit                                                     |
| Agriculture      | Crop planning (land, labor)                                                        |
| Diet Planning    | Minimum cost diet meeting nutrition                                                |
| Transportation   | Shortest/cheapest routes                                                           |
| Investment       | Portfolio optimization                                                             |
| Nigerian Context | Farmer Musa (rice vs cassava), Mama Ngozi (market goods), tailor (agbada vs suits) |

### **Nigerian Context Examples:**

**Farmer Musa: Should he plant rice or cassava to maximize profit?**

**Mama Ngozi: What mix of market goods gives maximum profit?**

**Tailor: Which combination of agbada and suits maximizes daily profit?**

**School: How to allocate budget between science and art equipment?**

### **WORKED EXAMPLES (Step-by-Step)**

#### **Example 1: Manufacturing Problem**





**A factory produces two types of chairs: dining chairs and office chairs. Each dining chair requires 2 hours of labor and 3 kg of wood. Each office chair requires 3 hours of labor and 2 kg of wood. The factory has 120 hours of labor and 150 kg of wood available per week. The profit per dining chair is ₦5,000 and per office chair is ₦6,000. Formulate this as a linear programming problem.**

**Solution:**

**Step 1: Define Decision Variables**

**Let  $x$  = number of dining chairs produced per week**

**Let  $y$  = number of office chairs produced per week**

**Step 2: Write Objective Function**

**Maximize Profit:  $P = 5000x + 6000y$**

**Step 3: Identify Constraints**

**Labor constraint:  $2x + 3y \leq 120$**

**Wood constraint:  $3x + 2y \leq 150$**

**Step 4: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 5: Complete Formulation**

**Maximize  $P = 5000x + 6000y$**

**Subject to:**



$$2x + 3y \leq 120$$

$$3x + 2y \leq 150$$

$$x \geq 0, y \geq 0$$

**Answer: The LP problem is formulated as above.**

### **Example 2: Farming Problem (Nigerian Context)**

**A farmer has 100 hectares of land. He can plant rice or maize. Rice requires 2 workers per hectare and maize requires 1 worker per hectare. He has 150 workers available. Rice yields a profit of ₦80,000 per hectare and maize yields ₦50,000 per hectare. Formulate this as a linear programming problem.**

**Solution:**

#### **Step 1: Define Decision Variables**

**Let  $x$  = number of hectares planted with rice**

**Let  $y$  = number of hectares planted with maize**

#### **Step 2: Write Objective Function**

**Maximize Profit:  $P = 80,000x + 50,000y$**

#### **Step 3: Identify Constraints**

**Land constraint:  $x + y \leq 100$**

**Labor constraint:  $2x + y \leq 150$**





**Step 4: Add Non-negativity**

$$x \geq 0, y \geq 0$$

**Step 5: Complete Formulation**

$$\text{Maximize } P = 80,000x + 50,000y$$

**Subject to:**

$$x + y \leq 100$$

$$2x + y \leq 150$$

$$x \geq 0, y \geq 0$$

**Answer: The LP problem is formulated as above.**

**Example 3: Minimization Problem (Diet Planning)**

**A nutritionist is planning a diet that must contain at least 300 calories and at least 50g of protein. Food A provides 100 calories and 20g of protein per serving, costing ₦200. Food B provides 150 calories and 10g of protein per serving, costing ₦250. Formulate to minimize cost.**

**Solution:**

**Step 1: Define Decision Variables**

**Let  $x$  = number of servings of Food A**

**Let  $y$  = number of servings of Food B**



**Step 2: Write Objective Function**

**Minimize Cost:  $C = 200x + 250y$**

**Step 3: Identify Constraints**

**Calorie requirement:  $100x + 150y \geq 300$  (at least 300 calories)**

**Protein requirement:  $20x + 10y \geq 50$  (at least 50g protein)**

**Step 4: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 5: Complete Formulation**

**Minimize  $C = 200x + 250y$**

**Subject to:**

**$100x + 150y \geq 300$**

**$20x + 10y \geq 50$**

**$x \geq 0, y \geq 0$**



**Answer: The LP problem is formulated as above.**

**Example 4: Tailor Problem (Nigerian Context)**

**A tailor makes traditional wear (agbada) and modern suits. Making an agbada takes 4 hours and uses 6 meters of fabric. Making a suit takes 3 hours and uses 4 meters of fabric. The tailor works 60 hours per**



week and has 90 meters of fabric. Profit is ₦15,000 per agbada and ₦12,000 per suit. Formulate this problem.

**Solution:**

**Step 1: Define Decision Variables**

**Let  $x$  = number of agbada made per week**

**Let  $y$  = number of suits made per week**

**Step 2: Write Objective Function**

**Maximize Profit:  $P = 15,000x + 12,000y$**

**Step 3: Identify Constraints**

**Time constraint:  $4x + 3y \leq 60$**

**Fabric constraint:  $6x + 4y \leq 90$**

**Step 4: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 5: Complete Formulation**

**Maximize  $P = 15,000x + 12,000y$**

**Subject to:**

**$4x + 3y \leq 60$**

**$6x + 4y \leq 90$**



$$x \geq 0, y \geq 0$$

**Answer: The LP problem is formulated as above.**

### **Example 5: Identify Linear Programming Problems**

**Which of the following are linear programming problems?**

**(a) Maximize  $P = 3x + 2y$ , subject to  $x^2 + y \leq 10$ ,  $x \geq 0$ ,  $y \geq 0$**

**Step 1: Check if all equations are linear**

**The constraint  $x^2 + y \leq 10$  contains  $x^2$  (non-linear term)**

**Answer: NOT Linear Programming because it contains  $x^2$**

**(b) Minimize  $C = 5x + 4y$ , subject to  $2x + 3y \geq 12$ ,  $x + y \leq 8$ ,  $x \geq 0$ ,  $y \geq 0$**

**Step 1: Check if all equations are linear**

**$2x + 3y \geq 12$  is linear**

**$x + y \leq 8$  is linear**

**$x \geq 0$ ,  $y \geq 0$  are linear**

**Answer: IS Linear Programming (all equations are linear)**



**(c) Maximize  $P = xy$ , subject to  $x + y \leq 20$ ,  $x \geq 0$ ,  $y \geq 0$**

**Step 1: Check if objective function is linear**

**$P = xy$  contains  $xy$  (non-linear term multiplication of variables)**

**Answer: NOT Linear Programming because objective function contains  $xy$**

### **CLASS ACTIVITIES (5–7 minutes)**

#### **Activity 1 (Group Work):**

**A school wants to buy computers and projectors. Computers cost ₦150,000 each and projectors cost ₦80,000 each. They have a budget of ₦2,000,000. Identify the decision variables and write the budget constraint.**

#### **Solution Guide:**

**Let  $x$  = number of computers**

**Let  $y$  = number of projectors**

**Budget constraint:  $150,000x + 80,000y \leq 2,000,000$**

#### **Activity 2 (Pair Work):**

**A bus company has 10 buses. Each bus needs 2 drivers and 1 mechanic per trip. There are 15 drivers and 8 mechanics available. Write the constraints.**

#### **Solution Guide:**



**Let  $x$  = number of buses used**

**Driver constraint:  $2x \leq 15$**

**Mechanic constraint:  $x \leq 8$**

**Bus availability:  $x \leq 10$**

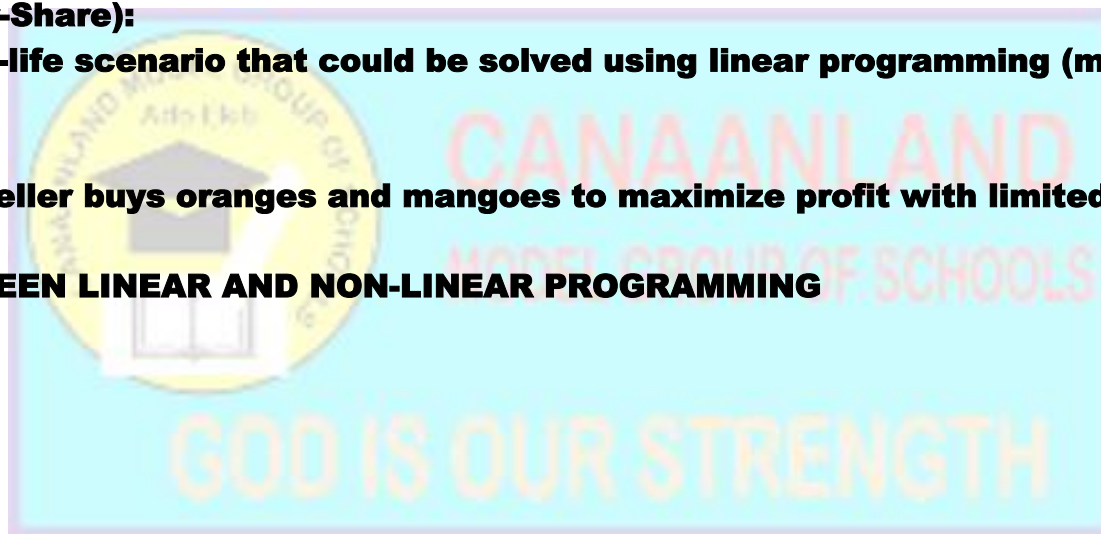
**Non-negativity:  $x \geq 0$**

**Activity 3 (Think-Pair-Share):**

**Create your own real-life scenario that could be solved using linear programming (market, farm, school, business).**

**Example: A market seller buys oranges and mangoes to maximize profit with limited money and space.**

**DIFFERENCES BETWEEN LINEAR AND NON-LINEAR PROGRAMMING**





| Feature                                                                                                   | Linear Programming (LP)                                                                      | Non-Linear Programming (NLP)                                                                                   |
|-----------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|  Variables               | First power only (x, y, z)<br>→ No exponents, no products                                    | Higher powers / products<br>→ $x^2$ , xy, $\sqrt{x}$ , $x^3$ , etc.                                            |
|  Graph shape             | ✓ <b>Straight lines</b> (linear equations/inequalities)<br>→ Planes in higher dimensions     | ✗ <b>Curves</b> — parabolas, circles, hyperbolas, exponential shapes                                           |
|  Ease of solution        | ● <b>Easy and efficient</b><br>→ Graphical method, Simplex algorithm, widely available tools | ● <b>Difficult</b><br>→ Requires advanced calculus, iterative methods (gradient descent, Lagrange multipliers) |
|  Real-world Example      | $2x + 3y \leq 100$<br>Maximize $P = 5x + 4y$                                                 | $x^2 + y \leq 100$ or $xy = 50$<br>(production with diminishing returns)                                       |
|  Optimal solution        | Always at a <b>corner point</b> (vertex) of feasible region                                  | Could be at boundary, interior, or multiple local optima                                                       |
|  Mathematical complexity | Linear algebra + simple inequalities                                                         | Calculus, convex analysis, global optimization techniques                                                      |

## QUICK SUMMARY FOR REVISION

**Linear Programming (LP) finds the best outcome with limited resources.**

**Decision variables are what you choose (x, y).**

**Objective function is what you maximize (profit) or minimize (cost).**

**Constraints are limitations (inequalities like  $\leq$  or  $\geq$ ).**

**Non-negativity means  $x \geq 0$ ,  $y \geq 0$  (no negative quantities).**



**Feasible region is all possible solutions.**

**Optimal solution is the best answer (highest profit or lowest cost).**

**LP is used in manufacturing, farming, diet planning, transportation, and investment.**

### **EVALUATION QUESTIONS (Classwork - Try These)**

**Question 1: Define linear programming and state its main components.**

**Answer: Linear Programming (LP) is a mathematical method for finding the best possible outcome (maximum profit or minimum cost) when there are limitations (constraints). The main components are: decision variables, objective function, constraints, non-negativity, feasible region, and optimal solution.**

**Question 2: What is the difference between an objective function and a constraint?**

**Answer: An objective function is what we want to maximize (profit) or minimize (cost). A constraint is a limitation or restriction on resources (time, money, materials). The objective function is the goal; constraints are the rules we must follow.**

**Question 3: A company makes products X and Y. Product X takes 2 hours and product Y takes 3 hours. There are 120 hours available. Write the time constraint.**

**Answer:  $2x + 3y \leq 120$**



**Question 4: State three real-life applications of linear programming.**

**Answer:**

**Manufacturing (product mix to maximize profit)**

**Agriculture (crop planning with limited land and labor)**

**Diet planning (cheapest food combination meeting nutrition needs)**

**Question 5: Why must decision variables in most LP problems be non-negative ( $x \geq 0$ ,  $y \geq 0$ )?**

**Answer: Because you cannot produce negative quantities of a product. For example, you cannot make -5 chairs. Physical quantities like number of products, hectares of land, or hours of work cannot be negative.**

**ASSIGNMENT (Homework)**

**Solve the following questions using the step-by-step method.**

**Question 1: Formulate this LP problem.**

**A farmer plants tomatoes and peppers. Tomatoes need 3 hours of labor per hectare and peppers need 2 hours of labor per hectare. He has 60 hours of labor available per week. Tomatoes give ₦40,000 profit per**



hectare and peppers give ₦30,000 profit per hectare. He has 25 hectares of land. Formulate this as an LP problem (define variables, objective function, and all constraints).

**Solution:**

**Step 1: Define Decision Variables**

**Let  $x$  = number of hectares planted with tomatoes**

**Let  $y$  = number of hectares planted with peppers**

**Step 2: Write Objective Function**

**Maximize Profit:  $P = 40,000x + 30,000y$**

**Step 3: Identify Constraints**

**Land constraint:  $x + y \leq 25$**

**Labor constraint:  $3x + 2y \leq 60$**

**Step 4: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 5: Complete Formulation**

**Maximize  $P = 40,000x + 30,000y$**

**Subject to:**

**$x + y \leq 25$**

**$3x + 2y \leq 60$**



$$x \geq 0, y \geq 0$$

**Answer: Formulated as above.**

**Question 2: Explain the meaning of the following terms in linear programming:**

**(a) Decision variables**

**Answer: The things we can control or choose (e.g., number of products to make, hectares to plant).**

**(b) Feasible solution**

**Answer: Any solution that satisfies all constraints (can be plotted inside the feasible region).**

**(c) Optimal solution**

**Answer: The best feasible solution that gives the highest profit or lowest cost.**

**(d) Constraint**

**Answer: A limitation or restriction on resources (time, money, materials, labor).**

**(e) Objective function**

**Answer: The formula that shows what we want to maximize (profit) or minimize (cost).**

**Question 3: Which of these are linear programming problems? Explain why or why not.**



**(a) Maximize  $P = 4x + 5y$ , subject to  $2x + 3y \leq 100$ ,  $x \geq 0$ ,  $y \geq 0$**

**Answer: IS Linear Programming because all equations are linear (variables to first power only, no  $x^2$ , no  $xy$ ).**

**(b) Maximize  $P = x^2 + y$ , subject to  $x + y \leq 50$ ,  $x \geq 0$ ,  $y \geq 0$**

**Answer: NOT Linear Programming because the objective function contains  $x^2$  (non-linear term).**

**(c) Minimize  $C = 3x + 7y$ , subject to  $x + 2y \geq 20$ ,  $3x + y \geq 30$ ,  $x \geq 0$ ,  $y \geq 0$**

**Answer: IS Linear Programming because all equations and the objective function are linear.**

**Question 4: Formulate to maximize profit.**

**A baker makes meat pies and sausage rolls. Each meat pie requires 2 eggs and 100g flour, costing ₦300 to make and selling for ₦500. Each sausage roll requires 1 egg and 50g flour, costing ₦150 to make and selling for ₦300. The baker has 100 eggs and 5000g (5kg) of flour available daily. Formulate this problem to maximize profit.**

**Solution:**

**Step 1: Define Decision Variables**

**Let  $x$  = number of meat pies made per day**





**Let  $y$  = number of sausage rolls made per day**

**Step 2: Calculate Profit per item**

**Meat pie profit = Selling price - Cost = 500 - 300 = ₦200**

**Sausage roll profit = 300 - 150 = ₦150**

**Step 3: Write Objective Function**

**Maximize Profit:  $P = 200x + 150y$**

**Step 4: Identify Constraints**

**Egg constraint:  $2x + y \leq 100$**

**Flour constraint:  $100x + 50y \leq 5000$  (divide by 50 to simplify:  $2x + y \leq 100$ )**

**Step 5: Add Non-negativity**

**$x \geq 0, y \geq 0$**

**Step 6: Complete Formulation**

**Maximize  $P = 200x + 150y$**

**Subject to:**

**$2x + y \leq 100$**

**$x \geq 0, y \geq 0$**

**Answer: Formulated as above.**



**Question 5: Give two examples of situations in your community or school that could be modeled using linear programming.**

**Answer:**

**Example 1 (School): A school wants to buy textbooks and exercise books with a limited budget to maximize the number of students served.**

**Let  $x$  = number of textbooks,  $y$  = number of exercise books**

**Maximize number of students served**

**Constraints: budget limit, storage space**

**Example 2 (Market): A market seller has limited money and space to buy yams and cassava to maximize profit.**

**Let  $x$  = number of yams,  $y$  = number of cassava**

**Maximize profit**

**Constraints: capital (money), space in shop, demand**

**SUMMARY FOR BOARD/DISPLAY (Key Points)**

|                                                          |         |
|----------------------------------------------------------|---------|
| +-----+-----+                                            |         |
| Term                                                     | Meaning |
| +-----+-----+                                            |         |
| Linear Programming   Best outcome with limited resources |         |



| **Decision Variables** | What we choose (x, y) |  
| **Objective Function** | What we maximize (profit) or minimize (cost) |  
| **Constraints** | Limitations (inequalities like  $\leq$  or  $\geq$ ) |  
| **Non-negativity** |  $x \geq 0, y \geq 0$  |  
| **Feasible Region** | All possible solutions (shaded area) |  
| **Optimal Solution** | Best answer (corner point) |

+-----+

### **GOLDEN RULE FOR FORMULATION**

**Read the problem carefully.**

**Define x and y clearly (what does each represent?).**

**Write the objective (Maximize or Minimize).**

**List all constraints from the numbers given.**

**Always add  $x \geq 0, y \geq 0$ .**

**END OF LESSON NOTE**